SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Revision Date 20-Jun-2024 **Revision Number** 4

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	**Karl Fischer Composite K2, for volumetric one-component titration, for ketones and aldehydes**
Cat No. :	**47159**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

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Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

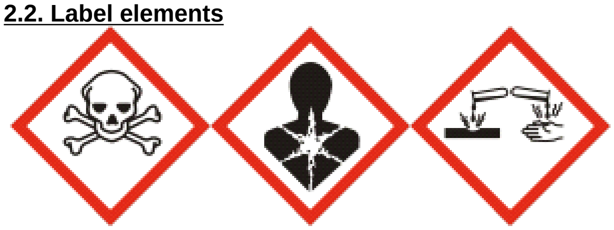
Acute Inhalation Toxicity - Vapors	Category 3 (H331)
Skin Corrosion/Irritation	Category 1 (H314) B
Serious Eye Damage/Eye Irritation	Category 1 (H318)
Reproductive Toxicity	Category 1B (H360D)
Specific target organ toxicity - (single exposure)	Category 1 (H370)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word **Danger**

Hazard Statements

H331 - Toxic if inhaled
H314 - Causes severe skin burns and eye damage
H370 - Causes damage to organs
H360D - May damage the unborn child

Precautionary Statements

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician

Additional EU labelling

Restricted to professional users

2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors
Toxic to terrestrial vertebrates

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
1-Imidazole	288-32-4	EEC No. 206-019-2	15.0	Acute Tox. 4 (H302) Skin Corr. 1C (H314) Eye Dam. 1 (H318) Repr. 1B (H360D)
Sulfur dioxide	7446-09-5	EEC No. 231-195-2	12.5	Press. Gas (H280) Skin Corr. 1B (H314) Eye Dam. 1 (H318) STOT SE 1 (H370) Acute Tox. 3 (H331)
Iodine	7553-56-2	231-442-4	10.0	Acute Tox. 4 (H332) Acute Tox. 4 (H312) Aquatic Acute 1 (H400)

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice
Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Eye Contact
Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

Skin Contact
Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Ingestion
Do NOT induce vomiting. Call a physician or poison control center immediately.

Inhalation
If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

Self-Protection of the First Aider
No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician
Treat symptomatically. Symptoms may be delayed.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
CO_2, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

Hazardous Combustion Products
Carbon oxides, Nitrogen oxides (NOx), Sulfur oxides, Hydrogen iodide.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class/LGK 6.1C

Storage Class (LGK) (Germany)

Switzerland - Storage of hazardous substances

Storage class - SC 6.1
https://www.kvu.ch/de/themen/stoffe-und-produkte
https://www.kvu.ch/fr/themes/substances-et-produits
https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

Component	European Union	The United Kingdom	France	Belgium	Spain
Sulfur dioxide	TWA: 1.3 mg/m³ (8h) TWA: 0.5 ppm (8h) STEL: 2.7 mg/m³ (15min) STEL: 1 ppm (15min)	STEL: 1 ppm 15 min STEL: 2.7 mg/m³ 15 min TWA: 0.5 ppm 8 hr TWA: 1.3 mg/m³ 8 hr	TWA / VME: 0.5 ppm (8 heures). TWA / VME: 1.3 mg/m³ (8 heures). STEL / VLCT: 1 ppm. indicative limit STEL / VLCT: 2.7 mg/m³. indicative limit	TWA: 0.5 ppm 8 uren TWA: 1.3 mg/m³ 8 uren STEL: 1 ppm 15 minuten STEL: 2.7 mg/m³ 15 minuten	STEL / VLA-EC: 2 ppm (15 minutos). STEL / VLA-EC: 5.28 mg/m³ (15 minutos). TWA / VLA-ED: 0.5 ppm (8 horas) TWA / VLA-ED: 1.32 mg/m³ (8 horas)
Iodine		STEL: 0.1 ppm; 1.1mg/m³	STEL / VLCT: 0.1 ppm. STEL / VLCT: 1 mg/m³.	TWA 0.1ppm; TWA 1mg/m³	STEL / VLA-EC: 0.1 ppm (15 minutos). STEL / VLA-EC: 1 mg/m³ (15 minutos). TWA / VLA-ED: 0.01 ppm (8 horas) TWA / VLA-ED: 0.1 mg/m³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Sulfur dioxide	TWA: 1.3 mg/m³ 8 ore. Time Weighted Average TWA: 0.5 ppm 8 ore. Time Weighted Average STEL: 2.7 mg/m³ 15 minuti. Short-term STEL: 1 ppm 15 minuti. Short-term	TWA: 1 ppm TWA: 2.5 mg/m³	STEL: 1 ppm 15 minutos STEL: 2.7 mg/m³ 15 minutos TWA: 0.5 ppm 8 horas TWA: 1.3 mg/m³ 8 horas	STEL: 0.7 mg/m³ MAC: 2 ppm MAC: 5 mg/m³	TWA: 0.5 ppm 8 tunteina TWA: 1.3 mg/m³ 8 tunteina STEL: 1 ppm 15 minuutteina STEL: 2.7 mg/m³ 15 minuutteina
Iodine		TWA: 0.1 ppm TWA: 1.1 mg/m³ skin absorber	STEL: 0.1 ppm 15 minutos TWA: 0.01 ppm 8 horas	0.1ppm MAC; 1mg/m³ MAC	STEL: 0.1 ppm 15 minuutteina STEL: 1.1 mg/m³ 15 minuutteina Iho

Component	Austria	Denmark	Switzerland	Poland	Norway
Sulfur dioxide	MAK-KZGW: 1 ppm 15 Minuten MAK-KZGW: 2.7 mg/m³ 15 Minuten MAK-TMW: 0.5 ppm 8 Stunden	TWA: 0.5 ppm 8 timer TWA: 1.3 mg/m³ 8 timer STEL: 2.7 mg/m³ 15 minutter STEL: 1 ppm 15 minutter	STEL: 1 ppm 15 Minuten STEL: 2.7 mg/m³ 15 Minuten TWA: 0.5 ppm 8 Stunden	STEL: 2.7 mg/m³ 15 minutach TWA: 1.3 mg/m³ 8 godzinach	TWA: 0.5 ppm 8 timer TWA: 1.3 mg/m³ 8 timer STEL: 1 ppm 15 minutter. value from the regulation STEL: 2.7 mg/m³ 15

	MAK-TMW: 1.3 mg/m³ 8 Stunden		TWA: 1.3 mg/m³ 8 Stunden		minutter. value from the regulation
Iodine	Haut MAK-KZGW: 0.1 ppm 15 Minuten MAK-KZGW: 1 mg/m³ 15 Minuten MAK-TMW: 0.1 ppm 8 Stunden MAK-TMW: 1 mg/m³ 8 Stunden Ceiling: 0.1 ppm Ceiling: 1 mg/m³	Ceiling: 0.1 ppm Ceiling: 1 mg/m³	Haut/Peau STEL: 0.1 ppm 15 Minuten STEL: 1 mg/m³ 15 Minuten TWA: 0.1 ppm 8 Stunden TWA: 1 mg/m³ 8 Stunden	STEL: 1 mg/m³ 15 minutach TWA: 0.5 mg/m³ 8 godzinach	Ceiling: 0.1 ppm Ceiling: 1 mg/m³

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Sulfur dioxide	TWA: 1.3 mg/m³ TWA: 0.5 ppm STEL : 2.7 mg/m³ STEL : 1 ppm	TWA-GVI: 0.5 ppm 8 satima. TWA-GVI: 1.3 mg/m³ 8 satima. STEL-KGVI: 1 ppm 15 minutama. STEL-KGVI: 2.7 mg/m³ 15 minutama.	TWA: 0.5 ppm 8 hr. TWA: 1.3 mg/m³ 8 hr. STEL: 2.7 mg/m³ 15 min STEL: 1 ppm 15 min	STEL: 2.7 mg/m³ STEL: 1 ppm TWA: 1.3 mg/m³ TWA: 0.5 ppm	TWA: 1.3 mg/m³ 8 hodinách. Ceiling: 2.7 mg/m³
Iodine	TWA: 3.0 mg/m³	STEL-KGVI: 0.1 ppm 15 minutama. STEL-KGVI: 1.1 mg/m³ 15 minutama.	TWA: 0.01 ppm 8 hr. inhalable fraction and vapour TWA: 0.01 mg/m³ 8 hr. STEL: 0.1 ppm 15 min		TWA: 0.1 mg/m³ 8 hodinách. Ceiling: 1 mg/m³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Sulfur dioxide	TWA: 0.5 ppm 8 tundides. TWA: 1.3 mg/m³ 8 tundides. STEL: 1 ppm 15 minutites. STEL: 2.7 mg/m³ 15 minutites.	TWA: 1.3 mg/m³ 8 hr TWA: 0.5 ppm 8 hr STEL: 2.7 mg/m³ 15 min STEL: 1 ppm 15 min	STEL: 1 ppm STEL: 2.7 mg/m³ TWA: 0.5 ppm TWA: 1.3 mg/m³	STEL: 2.7 mg/m³ 15 percekben. CK STEL: 1 ppm 15 percekben. CK TWA: 1.3 mg/m³ 8 órában. AK TWA: 0.5 ppm 8 órában. AK	STEL: 1 ppm STEL: 2.7 mg/m³ TWA: 0.5 ppm 8 klukkustundum. TWA: 1.3 mg/m³ 8 klukkustundum.
Iodine	STEL: 0.1 ppm 15 minutites. STEL: 1 mg/m³ 15 minutites.		STEL: 0.1 ppm STEL: 1 mg/m³ TWA: 0.1 ppm TWA: 1 mg/m³	STEL: 1 mg/m³ 15 percekben. CK STEL: 0.1 ppm 15 percekben. CK TWA: 1 mg/m³ 8 órában. AK TWA: 0.1 ppm 8 órában. AK lehetséges borön keresztüli felszívódás	STEL: 0.1 ppm STEL: 1 mg/m³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Sulfur dioxide	STEL: 2.7 mg/m³ STEL: 1 ppm TWA: 1.3 mg/m³ TWA: 0.5 ppm	TWA: 1.3 mg/m³ IPRD TWA: 0.5 ppm IPRD STEL: 2.7 mg/m³ STEL: 1 ppm	TWA: 1.3 mg/m³ 8 Stunden TWA: 0.5 ppm 8 Stunden STEL: 2.7 mg/m³ 15 Minuten STEL: 1 ppm 15 Minuten	TWA: 0.5 ppm TWA: 1.3 mg/m³ STEL: 1 ppm 15 minuti STEL: 2.7 mg/m³ 15 minuti	TWA: 0.5 ppm 8 ore TWA: 1.3 mg/m³ 8 ore STEL: 1 ppm 15 minute STEL: 2.7 mg/m³ 15 minute
Iodine	TWA: 1 mg/m³	Ceiling: 0.1 ppm Ceiling: 1 mg/m³			TWA: 0.09 ppm 8 ore TWA: 0.5 mg/m³ 8 ore STEL: 0.2 ppm 15 minute STEL: 1 mg/m³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Sulfur dioxide	Skin notation	Ceiling: 2.7 mg/m³	TWA: 0.5 ppm 8 urah	Binding STEL: 1 ppm 15	

	MAC: 10 mg/m^3	TWA: 0.5 ppm TWA: 1.3 mg/m^3	TWA: 1.3 mg/m^3 8 urah STEL: 1 ppm 15 minutah STEL: 2.7 mg/m^3 15 minutah	minuter Binding STEL: 2.7 mg/m^3 15 minuter TLV: 0.5 ppm 8 timmar. NGV TLV: 1.3 mg/m^3 8 timmar. NGV	
Iodine	Skin notation MAC: 1 mg/m^3	Ceiling: 1.1 mg/m^3 TWA: 0.1 ppm TWA: 1.1 mg/m^3		Binding STEL: 0.1 ppm 15 minuter Binding STEL: 1 mg/m^3 15 minuter	

Biological limit values
This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)
See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
1-Imidazole 288-32-4 (15.0)				DNEL = 1.5mg/kg bw/day
Iodine 7553-56-2 (10.0)				DNEL = 0.01mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
1-Imidazole 288-32-4 (15.0)				DNEL = 10.6mg/m^3
Sulfur dioxide 7446-09-5 (12.5)	DNEL = 2.7mg/m^3		DNEL = 2.7mg/m^3	
Iodine 7553-56-2 (10.0)				DNEL = 0.07mg/m^3

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
1-Imidazole 288-32-4 (15.0)	PNEC = 0.13mg/L	PNEC = 0.336mg/kg sediment dw	PNEC = 1.3mg/L	PNEC = 10mg/L	PNEC = 0.0425mg/kg soil dw
Iodine 7553-56-2 (10.0)	PNEC = 18.13µg/L	PNEC = 3.99mg/kg sediment dw		PNEC = 11mg/L	PNEC = 5.95mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
1-Imidazole	PNEC = 0.013mg/L	PNEC =			

288-32-4 (15.0)		0.0336mg/kg sediment dw			
Iodine 7553-56-2 (10.0)	PNEC = 60.01µg/L	PNEC = 20.22mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures
None under normal use conditions. Ensure that eyewash stations and safety showers are close to the workstation location.

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Viton (R)	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection No protective equipment is needed under normal use conditions.

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Particle filter

Small scale/Laboratory use Maintain adequate ventilation
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

Environmental exposure controls Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State Liquid

Appearance Dark brown
Odor No information available
Odor Threshold No data available
Melting Point/Range No data available
Softening Point No data available
Boiling Point/Range No information available
Flammability (liquid) No data available
Flammability (solid,gas) Not applicable Liquid
Explosion Limits No data available

Flash Point No information available **Method -** No information available

Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	Not applicable	
Viscosity	No data available	
Water Solubility	No information available	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
1-Imidazole	-0.02	
Iodine	2.49	
Vapor Pressure	No data available	
Density / Specific Gravity	No data available	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

Carbon oxides. Nitrogen oxides (NOx). Sulfur oxides. Hydrogen iodide.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral	Based on available data, the classification criteria are not met
Dermal	Based on available data, the classification criteria are not met
Inhalation	Category 3

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation

1-Imidazole	970 mg/kg (Rat)	-	-
Sulfur dioxide	-	-	Per CGA P-20: 2500 ppm/1hr (Rat)
Iodine	315 mg/kg (Rat)	1425 mg/kg (Rabbit)	4.588 mg/L 4h (Rat)

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;
Respiratory No data available
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Category 1B

(h) STOT-single exposure; Category 1

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects,both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects The product contains following substances which are hazardous for the environment. Contains a substance which is:. Very toxic to aquatic organisms.

Component	Freshwater Fish	Water Flea	Freshwater Algae
1-Imidazole		EC50: = 341.5 mg/L, 48h (Daphnia magna)	EC50: = 82 mg/L, 96h (Desmodesmus subspicatus) EC50: = 130 mg/L, 72h (Desmodesmus subspicatus)
Iodine	Oncorhynchus mykiss: LC50 =	EC50 = 0,2 mg/l/48 h	-

	1,7 mg/l/96 h		

Component	Microtox	M-Factor
1-Imidazole	= 1200 mg/L EC50 Pseudomonas putida 17 h = 231 mg/L EC50 Photobacterium phosphoreum 30 min	
Iodine	-	

12.2. Persistence and degradability No information available

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential No information available

Component	log Pow	Bioconcentration factor (BCF)
1-Imidazole	-0.02	No data available
Iodine	2.49	No data available

12.4. Mobility in soil No information available

12.5. Results of PBT and vPvB assessment No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC) According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

Switzerland - Waste Ordinance Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number UN3267
14.2. UN proper shipping name CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.
Technical Shipping Name Sulfur dioxide/1-imidazole
14.3. Transport hazard class(es) 8
14.4. Packing group III

ADR

14.1. UN number UN3267
14.2. UN proper shipping name CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.
Technical Shipping Name Sulfur dioxide/1-imidazole
14.3. Transport hazard class(es) 8
14.4. Packing group III

IATA

14.1. UN number UN3267
14.2. UN proper shipping name CORROSIVE LIQUID, BASIC, ORGANIC, N.O.S.
Technical Shipping Name Sulfur dioxide/1-imidazole
14.3. Transport hazard class(es) 8
14.4. Packing group III

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
1-Imidazole	288-32-4	206-019-2	-	-	X	X	KE-20937	X	X
Sulfur dioxide	7446-09-5	231-195-2	-	-	X	X	KE-32567	X	X
Iodine	7553-56-2	231-442-4	-	-	X	X	KE-21023	X	-

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
1-Imidazole	288-32-4	X	ACTIVE	X	-	X	X	X
Sulfur dioxide	7446-09-5	X	ACTIVE	X	-	X	X	X
Iodine	7553-56-2	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) -	REACH (1907/2006) -	REACH Regulation (EC

		Annex XIV - Substances Subject to Authorization	Annex XVII - Restrictions on Certain Dangerous Substances	1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
1-Imidazole	288-32-4	-	Use restricted. See entry 30. (see link for restriction details) Use restricted. See entry 75. (see link for restriction details)	-
Sulfur dioxide	7446-09-5	-	Use restricted. See entry 75. (see link for restriction details)	-
Iodine	7553-56-2	-	Use restricted. See entry 75. (see link for restriction details)	-

REACH links
https://echa.europa.eu/substances-restricted-under-reach

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
1-Imidazole	288-32-4	Not applicable	Not applicable
Sulfur dioxide	7446-09-5	Not applicable	Not applicable
Iodine	7553-56-2	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values
Take note of Directive 94/33/EC on the protection of young people at work
Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 2 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
1-Imidazole	WGK2	
Sulfur dioxide	WGK1	
Iodine	WGK 2	

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Iodine 7553-56-2 (10.0)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3
H331 - Toxic if inhaled
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H370 - Causes damage to organs
H360D - May damage the unborn child
H302 - Harmful if swallowed
H312 - Harmful in contact with skin
H332 - Harmful if inhaled
H400 - Very toxic to aquatic life

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Prepared By	Health, Safety and Environmental Department
Revision Date	20-Jun-2024
Revision Summary	New emergency telephone response service provider.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer
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End of Safety Data Sheet